Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/06/2008
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Project Title:	Fire Suppression and Safety in Reduced Gravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	02-OBPR-08
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA97G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This study, along with the numerical simulation work conducted by the Laboratory for Computational Physics & Fluid Dynamics at the Naval Research Laboratory (NRL), focuses on the suppression of small fires in large compartments and enclosures in various gravitational environments, such as the ones experienced by spacecraft in orbit and on the surface of the Moon and Mars. The primary goal of the experimental effort is to provide data, via ground-based, normal-gravity tests, for validating key elements of the numerical model, while providing complementary information on the interactions between the suppressant and the fire. Two classes of experiments are conducted: gaseous-fuel diffusion flames and solid-fuel thermoplastic fires. On both fire scenarios, fine water mist is used as the suppression agent. The experimental validation program provides the close coupling of the experiments to the numerical simulation work performed by the Naval Research Laboratory team (whose numerical results are published in a separate report), while simultaneously providing a realistic fire scenario in which many factors including gravity effects, obstructions, and ventilation, play a part in the response of the fire to the suppressant. In this way, the experimental work provides rapid		

	validation data for the models and a systematic method for approaching realistic water mist fire suppression design problems for applications typically protected by other means, thus enabling a reduction in the amount of full-scale testing needed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will also provide additional basic information on fire suppression and thus will enhance our knowledge base for fighting fires on Earth, specially fires on enclosed spaces such as aircraft cabins and cargo bays, ships, machinery and telecommunication rooms, museums, and libraries.
Task Progress:	The work performed to characterize the suppression effectiveness of water mist on gaseous-fuel diffusion flames is conducted on a 1.25-cm diameter methane gas-jet burner where information is gathered on flame heights, fuel flow, and fuel velocity. The first set of experiments consists of flowing methane (99.99%) through the tube at various flow rates varying from 0 to 300 sccm, corresponding to flow velocities of approximately 0-240 cm/min. It is noticed that the flame height grows linearly with flow velocity up to a velocity of approximately 240 cm/min, after which the flames become unstable and tend to flicker. Subsequent tests include an incursion into the unstable flow region to replicate flame conditions encountered in turbulent gas-jet flames, which correspond to typical flames found in most fire scenarios. Video images and temperature measurements are taken from flames with methane fuel flows of 250 to 1000 sccm ejecting into open air in the combustion chamber. After running the dry runs described above, several tests were conducted to characterize the suppression and extinction behavior of gaseous-fuel diffusion flames. Initially, a water-mist nozzle located 50 cm directly above the burner is selected with a 0.20-mm orifice and with a spray cone angle of 30 degrees. For all tests it is run at 100 psi and a water flow rate of 0.5 ml/s, which is the lowest flow rate possible for this nozzle. It is necessary to go this low in flow rate because the immediate suppression of all flames at higher flows. The Sauter Mean Diameter (D32) of the water mist for this nozzle at this pressure is approximately 35 microns as measured at a 15-cm distance from the tip of the nozzle. For a methane flow rate of 200 cc/min (with a steady flame), an average extinction time of 0.5 seconds is observed. The flame is very quickly extinguished as soon as it gets in contact with the water mist. The flame appears to broaden and move rapidly from side to side before extinction. Its temperature is quickly lowered by the mist as evidenced from the transition from a bright yellow flame to a dim orange-blue flame before total extinction. For a methane flow rate of 350 cc/min (with an unsteady, flickering flame), an average extinction time of 2.5 seconds is observed. For this higher methane flow rate, the unstable flame resists the flow of water mist for a longer time than the case of the smaller stable flame described above. As in the case of a lower methane flow rate, the flame appears to broaden as soon as it gets in contact with the mist, and moves rapidly from side to side before extinction. Similarly, the flame temperature is quickly lowered by the mist as evidenced from the transition from a bright yellow flame to a dim orange-blue flame before total extinction. In order to prolong the extinction time to observe the behavior of the flame in more detail, it was decided to locate the nozzle off-axis, but closer to the burner. Thus, a nozzle height of 35 cm above the flame and an off-axis distance of 2 cm are selected with a water flow rate of 0.65 ml/s. The extinction times turned out to be quite sensitive to the flow rate, so the flow was slowly increased until the extinction times consisted of a few seconds (to avoid long computer processing time during the simulations). For a methane flow rate of 200 cc/min (with a steady flame), an average extinction time of 5.4 seconds is observed. In the case of a methane flow rate of 350 cc/min (with an unsteady, flickering flame), an extinction time of 8.3 seconds is recorded. These measurements reflect an average result from 10 tests per case. These were the final numbers provided to the NRL team for their incorporation into the numerical model for validation and verification of the gaseous–fuel diffusion flame case. To complement the gaseous-fuel flame studies discussed above and the experimental results provided to NRL, an investigation on the effectiveness of water mist as a fire suppressant in a more conventional environment is undertaken. The goals of this part of the investigation are twofold: to provide a more realistic fire scenario in which many factors, including gravity effects, obstructions, and ventilation play a part in the response of the fire to the suppressant and to develop a systematic method for approaching realistic water mist fire suppression design problems for applications typically protected by other means, possibly enabling a reduction in the amount of full-scale testing needed. Data produced in this work can be used to calibrate future numerical models that may aid both the scientific community and fire engineers in better understanding water mist fire suppression. The system selected for study is water-mist suppression of real-scale thermoplastic fires in electronics cabinets. Water mist shouldn’t be expected to behave like a total flooding agent, however it can still be effective as a fire suppressant for electronics cabinets if the system is designed in a manner that allows the mist to penetrate into all flammable areas. The most effective nozzle configuration used in this work was a four-nozzle cluster, which incorporate the benefits of a direct spray, which can quickly lower the local temperature and oxygen levels below their thresholds capable of sustaining combustion, with an indirect spray that prevents the recovery of the local oxygen concentration and lowers the ambient temperature in the cabinet. Compared with tests of a single nozzle, this nozzle configuration confirms that the reduction of the local oxygen concentration is more important for fire extinction than lowering the local temperature. The optimal nozzle configuration (the four-nozzle cluster) for the experiments in this investigation allows relatively small droplets to penetrate to the base of the fire. If the droplets evaporate on or in the near vicinity of the burning surface the fire is extinguished immediately. This result indicates that the droplet momentum and focus of the spray is very important for water mist fire suppression in this application. The droplet size is important because, especially in this application, any excess water is highly undesirable. However, if the droplets cannot reach their most effective location, their performance suffers and the penalty of damaged equipment is high.
Bibliography Type:	Description: (Last Updated: 03/21/2008)
Papers from Meeting Proceedings	Schwer DA, Kailasanath K, Abbud-Madrid A. "Development of a Fire Suppression Simulation Tool for Various Low-Gravity Environments." 45th Aerospace Sciences Meeting & Exhibit, American Institute of Aeronautics and Astronautics, Reno, NV, January 8-11, 2007. 45th Aerospace Sciences Meeting & Exhibit, American Institute of Aeronautics and Astronautics, AIAA Paper 2007-0735, 2007. , Jan-2007